FU Xiao

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Education

Zhejiang University (ZJU) Hangzhou, China

Bachelor of Information Eng. | College of Information Science & Electronic Engineering (ISEE) *Sept. 2018 – Jul. 2022*

- **GPA:** 3.96/4.00, 90.63/100, **Ranking:** 1/144
- **Laureate of National Scholarships (twice), Graduated with Dean's Honor**
- **Honors Program:** ZJU Morningside Cultural China Scholars Program (36 students shortlisted from 12,000 freshmen with distinguished innovative capability and leadership)
- **Language Proficiency:** TOEFL (iBT) : 107 (R:29,L:28,S:22,W:28)

Publications

*** equal contribution, † corresponding author**

- Tong Wu, Jiarui Zhang, Xiao Fu, Yuxin Wang, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, Ziwei Liu
  "OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation"
  Under review of *Computer Vision and Pattern Recognition (CVPR)*, 2023, Vancouver, Canada (ID : 2642) [**Paper**]
- Xiao Fu*, Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao†
  "Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation"
  In *Proceedings of the International Conference on 3D Vision (3DV)*, 2022, Prague, Czechia, pp. 301-311, doi: 10.1109/3DV57658.2022.00042 [**Paper**|**Website**|**Poster**|**Video**|**Code**]
- Guangyi Zhang*, Xiao Fu*, Qiyu Hu†, Yunlong Cai, Guanding Yu
  "Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network"
  In *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2021, Helsinki, Finland, pp. 678-683, doi: 10.1109/PIMRC50174.2021.9569633 [**Paper**|**Slides**]

Research Experience

Neural Panoptic 3D Reconstruction of Large-Scale Real-World Scenario Shanghai AI Laboratory

Advisor: Prof. Dahua Lin and Dr. Jiaqi Wang *Aug. 2022 – Now*

- generated high-quality scene reconstruction and panoptic labels in 3D space from multi-view images and semantic label results predicted by 2D pre-trained network on the ScanNet dataset
- leveraged the coarse semantic segmentations of 2D images and the SDFs of scene reconstruction to obtain the accurate instance labels in 3D space, especially in overlapped regions of objects
- employed the reconstruction priors to improve the accuracy of 3D semantic/instance segmentation masks
- intending to publish a pertinent paper to ICCV'23 in capacity of the first author

OmniObject3D: A Large-Vocabulary and Realistic 3D Object Dataset Shanghai AI Laboratory

Advisor: Prof. *Dahua Lin* and Dr. *Tong Wu* *Jul. 2022 – Nov. 2022*

- proposed OmniObject3D, a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects
- (1) Large Vocabulary: comprises 6,000 scanned objects in 190 daily categories, sharing common classes with popular 2D datasets, benefiting the pursuit of generalizable 3D representations
- (2) Rich Annotations: each 3D object is captured with both 2D and 3D sensors, providing textured meshes, point clouds, multi-view rendered images, and multiple real-captured videos
- (3) Realistic Scans: provides professionally scanned object scans with precise shapes and realistic appearances
- set up four tracks: robust 3D perception, novel-view synthesis, neural surface reconstruction, and 3D generation

Unaligned Sparse Object-Centric Image Synthesis Shanghai AI Laboratory

Advisor: Dr. Jiaqi Wang and Dr. Sida Peng *Mar. 2022 – Jul. 2022*

- focused on view synthesis from sparse inputs on unaligned coordinate system, a more generic scenario, where category-level priors can not be stored by *xyz* inputs
- projected the features of source views in 3D space via monocular depth predicted by MiDaS and replaced *xyz*, from which network can implicitly learn priors, in NeRFormer with nearest features at novel viewpoints

- outperformed NeRFormer w/o *xyz* and ViewFormer in terms of novel view synthesis, and was comparable with NeRFormer in case of sparse views (1-3 views)

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer Max Planck Institute for Intelligent Systems & ZJU
Advisor: Prof. *Andreas Geiger* and Prof. *Yiyi Liao* *Sept. 2021 – Mar. 2022*

- proposed to utilize neural rendering as a tool to generate high-quality 2D panoptic segmentation annotations from a set of sparse images and easy-to-obtain coarse 3D bounding primitives and noisy 2D semantic predictions
- leveraged a novel dual formulation of the semantic fields to improve the underlying geometry reconstruction and resolve label ambiguity at the intersection region of the 3D bounding primitives
- reduced the label annotation time from 1.5 hours to 0.75 minute per frame in a single street scene
- achieved SOTA label transfer results in terms of accuracy and multi-view consistency on urban scenes of the KITTI-360 dataset and enabled rendering of multi-view and spatio-temporally consistent labels at novel viewpoints

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network ZJU
Advisor: Prof. *Guanding Yu* *Jun. 2020 – Feb. 2021*

- proposed a learning-based framework to address the high computational complexity problem of a convex optimization algorithm, the dual-layer iterative Penality Dual Decomposition (PDD) algorithm
- unfolded the PDD algorithm into a layer-wise structure whereby some trainable parameters were introduced into the places of the high-complexity operations, e.g., large-dimensional matrix inversions and iterative sub algorithms
- focused on solving the spectrum efficiency maximization problem for hybrid pre-coding architecture
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

Honors & Awards

- National Scholarship, Ministry of Education of P.R. China **(Top 0.2% across China, Twice)** *2020,2021*
- ISEE Excellent Student Award **(Top 5 students among ISEE 314 undergraduates)** *2021*
- ISEE Yuelun Alumni Scholarship **(Dean's Award for Academic Contest)** *2020*
- First-Class Scholarship for Academic Excellence of Zhejiang Univeristy **(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 3%)** *2019*
- Outstanding Graduation Thesis Award of Zhejiang University *2022*
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM) **(Top 7% out of 26,112 teams)** *2021*
- Gold Prize, 7^{th} "Internet Plus" College Student Innovation and Entrepreneurship Contest **(1^{st} author)** *2021*
- Gold Prize, 12^{th} "Challenge Cup" National College Student Business Plan Competition, National *2020*
- Grand Prize, 12^{th} "Challenge Cup" National College Student Business Plan Competition, Provincial *2020*
- Grand Prize, "Yongdian Cup" Innovation and Entrepreneurship Contest **(1^{st} author)** *2019*
- Second Prize, College Student Physical Innovation Competition (Theoretical) of Zhejiang Province *2020*
- Third Prize, 17^{th} "Challenge Cup" Extracurricular Academic Science and Technology Contest **(1^{st} author)** *2021*
- Outstanding Student Award of Zhejiang University *2019,2020,2021*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*

Extracurricular Activities

ZJU Volunteer Organization Hangzhou, China
Participant *Sept. 2018 – Jun. 2022*

- contributed with over 250 volunteer hours by participating in more than 30 volunteer services, such as the "Chunlin" Anti-Epidemic Online Teaching activity, Prince Bay guidance service, "Internet Plus" reception service, ZJU autumn sporting events, and hometown anti-epidemic activities
- **Laureate of ZJU Five-Star Volunteer, the highest honor awarded for ZJU volunteer participants**

ISEE Youth League Committee Hangzhou, China
Deputy Secretary *Sept. 2020 – Jul. 2021*

- assisted the league committee teachers in carrying out academic competitions as well as innovation and entrepreneurship competitions ("Internet Plus" and "Challenge Cup")
- participated in the sharing session of outstanding seniors of ISEE, and the counterparts of the 19 grade assembly, epidemic prevention, and employment quality analysis
- **Laureate of Excellent Member Award of ZJU Youth League Committee**

Extracurricular Skills

Computer: Python, PyTorch, TensorFlow, C/C++, MATLAB, Java, LaTex, Linux, Caffe.